Control Theory

Chap 5: Analysis and Synthesis of State-space System

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Objectives:

1. Can equal the system models in Laplace transfer function and state-space equation;
2. Understand the equal meanings of Observer and Controller;
3. Analyze the system stability and performance modelled by state-space equation;
4. Master how to design the gains of the state feedback controller;
5. Master how to design the observer and observer based Controller.



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Chap 5: Analysis & Synthesis of State-space System

1. Introduction
2. State-space Equation Model
3. State - space Representation of Transfer - function Systems
4. Solving the Time-invariant State Equation
5. Controllability and Observability of System
6. Pole Placement
7. Stability Analysis based on Lyapunov Criterion
8. Controller based on Observer
9. Chapter Conclusion

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Why State-space Model?

A typical transfer function with one zero and two poles can be described in (1),

$$G(s) = \frac{(s - z_1)}{(s - p_1)(s - p_2)} \tag{1}$$

if the value z_1 equals p_1, (1) can be expressed as

$$G(s) = \frac{1}{(s - p_2)} \tag{2}$$

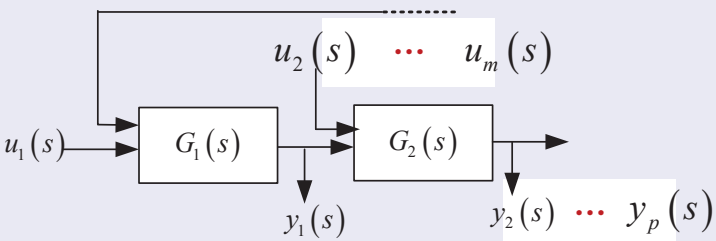


Figure 1: MIMO System

$$\begin{array}{cccc} \frac{Y_1(s)}{U_1(s)} = \cdots, & \frac{Y_2(s)}{U_1(s)} = \cdots, & \cdots., & \frac{Y_p(s)}{U_1(s)} = \cdots, \\ \frac{Y_1(s)}{U_2(s)} = \cdots, & \frac{Y_2(s)}{U_2(s)} = \cdots, & \cdots., & \frac{Y_p(s)}{U_2(s)} = \cdots, \\ \vdots & \vdots & \vdots & \vdots \\ \frac{Y_1(s)}{U_m(s)} = \cdots, & \frac{Y_2(s)}{U_m(s)} = \cdots, & \cdots., & \frac{Y_p(s)}{U_m(s)} = \cdots. \end{array} \tag{3}$$

State-space Model

$$\begin{cases} \dot{x}=Ax+Bu \\ y=Cx+Du \end{cases} \tag{4}$$

where $A \in \mathcal{R}^{n\times n}, B \in \mathcal{R}^{n\times m}, C \in \mathcal{R}^{p\times n}, D \in \mathcal{R}^{p\times m}$ $y \in \mathcal{R}^{p}$, $u \in \mathcal{R}^{m}$.

Table 1: Comparison between Conventional and Modern Control .

Item	Conventional Control	Modern Control
Domain	Time and Frequency Domain	Time Domain Directly
Math	Laplace transform	Linear algebra
Number of Input and Output	SISO	MIMO & SISO
Design Approaches	PID and Compensation	Pole placement, Eigenvalue assignment
Types of System	LTI	Nonlinear, Time-variable
Description Relationship	Input-output	Inside Description of System
Other		Easy on Designing Optimal Controller



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2 State-space Equation Model

- Electrical Circuits
- Euler-Lagrange Model

Consider following RLC electrical circuits

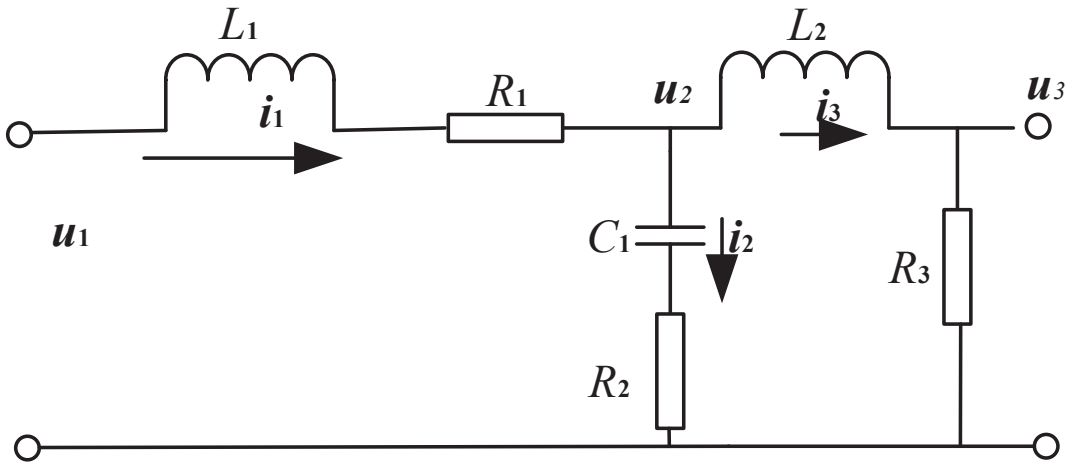


Figure 2: RLC Electrical Circuits

Via the Kirchoff's law, it can obtain that

$$\begin{aligned} u_2 - u_1 &= L_1\dot{i}_1 + i_1R_1 \\ i_2 &= C_1(\dot{u}_2 - \dot{i}_2R_2) \\ i_1 &= i_2 + i_3 \\ L_2\dot{i}_3 &= u_3 - u_2 \\ i_3R_3 &= u_3 \end{aligned}$$

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Choose the state variable i_1, u_2, and i_3 , it obtains that

$$\begin{aligned} L_1\dot{i}_1 &= u_2 - i_1R_1 - u_1 \\ C_1\dot{u}_2 - C_1R_2\dot{i}_1 + C_1R_2\dot{i}_3 &= i_1 - i_3 \\ L_2\dot{i}_3 &= i_3R_3 - u_2 \end{aligned}$$

and the output

$$u_3 = i_3R_3$$

such that the state space model can be obtained

$$\begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{cases} \tag{5}$$

where $u = u_1$, $y = u_3$ $A = \tilde{M}^{-1}\tilde{A}$, $B = \tilde{M}^{-1}\tilde{B}$,$C = [0\ 0\ R_3]$, $D = 0$,

$$x = \begin{bmatrix} i_1 \\ u_2 \\ i_3 \end{bmatrix}, \tilde{M} = \begin{bmatrix} L_1 & 0 & 0 \\ -C_1R_2 & C_1 & C_1R_2 \\ 0 & 0 & L_2 \end{bmatrix}, \tilde{A} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & R_3 \end{bmatrix}, \tilde{B} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$



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2 State-space Equation Model

- Electrical Circuits
- Euler-Lagrange Model

Consider an Euler-Lagrange used in robotics, aerospace and mechanics,

$$M(q)\ddot{q} = C(\dot{q}, q)\dot{q} + G(q) + \tau \tag{6}$$

$M(q)$	inertia matrix	$C(\dot{q}, q)$	centrifugal and Coriolis
G	gravity force	τ	control input

Present as state-space model below

$$\dot{x}(t) = Ax(t) + Bu \tag{7}$$

where $u = \tau(t)$,

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}, A = \begin{bmatrix} 0 & I \\ M(q)^{-1}\frac{\partial G(q)}{\partial q} & M(q)^{-1}C(q, \dot{q}) \end{bmatrix}, B = \begin{bmatrix} 0 \\ M(q)^{-1} \end{bmatrix}.$$

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General Type Differential Model

$$y^{(n)} + a_1 y^{(n-1)} + \ldots + a_{n-1} y^{(1)} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \ldots + b_{n-1} u^{(1)} + b_n u \tag{8}$$

y	system output
u	system input
$y^{(k)}$	k_{-th} order derivative of $y(t)$ $\frac{d^k y}{dt^k}$

The Laplacian transformation of (8) can be further depicted as

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^{(n)} + b_1 s^{(n-1)} + \ldots + b_{n-1} s^{(1)} + b_n}{s^{(n)} + a_1 s^{(n-1)} + \ldots + a_{n-1} s^{(1)} + a_n} \tag{9}$$

From State space to Transfer Function

$$G(s) = C(sI - A)^{-1} B + D$$

Not Only one Method from $G(s)$ to State space

$$\begin{cases} \dot{x} &= Ax(t) + Bu(t) \\ y &= Cx(t) + Du(t) \end{cases}$$

3 State - space Representation of Transfer - function Systems

- Controllable State-space Representation
- Observable State-space Representation
- Diagonal Canonical State-space Representation
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Represent the system in (8) and (9) into controllable state-space, where

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \dots & -a_2 & -a_1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix},$$
$$C = \begin{bmatrix} b_n - a_n b_0 & b_{n-1} - a_{n-1} b_0 & b_{n-2} - a_{n-1} b_0 & \dots & b_2 - a_2 b0 & b_1 - a_1 b0 \end{bmatrix}, D = b_0. \tag{10}$$

Proof.

Consider the transfer function in (9),

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s^1 + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s^1 + a_n} \tag{11}$$

and modify it into

$$Y(s) = b_0 U(s) + \hat{Y}(s) \tag{12}$$

where

$$\hat{Y}(s) = \frac{(b_1 - a_1 b_0) s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0) s + (b_n - a_n b_0)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} U(s)$$

□

Proof.

Define that

$$\begin{aligned} Q(s) &= \frac{\hat{Y}(s)}{(b_1 - a_1 b_0)s^{n-1} + \cdots + (b_{n-1} - a_{n-1}b_0)s + (b_n - a_n b_0)} \\ &= \frac{U(s)}{s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n} \end{aligned}$$

one has the following two equations that

$$\begin{aligned} s^n Q(s) &= -a_1 s^{n-1}Q(s) - \cdots - a_{n-1}sQ(s) - a_n Q(s) + U(s) \\ \hat{Y}(s) &= (b_1 - a_1 b_0)s^{n-1}Q(s) + \cdots \\ &\quad + (b_{n-1} - a_{n-1}b_0)sQ(s) + (b_n - a_n b_0)Q(s) \end{aligned} \tag{13}$$

Define the state variables as follows

$$\begin{aligned} X_1(s) &= Q(s) \\ X_2(s) &= sQ(s) \\ &\vdots \\ X_{n-1}(s) &= s^{n-2}Q(s) \\ X_n(s) &= s^{n-1}Q(s) \end{aligned}$$

□

Proof.

Clearly, it obtains that

$$\begin{aligned} sX_1(s) &= X_2(s) \\ sX_2(s) &= X_3(s) \\ &\vdots \\ sX_{n-1}(s) &= X_n(s) \end{aligned}$$

and can be written in time-domain as

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= x_3(t) \\ &\vdots \\ \dot{x}_{n-1}(t) &= x_n(t) \end{aligned} \tag{14}$$

and

$$\begin{aligned} sX_n(s) &= s^nQ(s) = -a_1s^{n-1}Q(s) - \cdots \\ &\quad -a_{n-1}sQ(s) - a_nQ(s) + U(s) \\ &= -a_1X_n(s) - \cdots - a_{n-1}X_2(s) - a_nX_1(s) + U(s) \end{aligned}$$

or

$$\dot{x}_n(t) = -a_1x_n(t) - \cdots - a_{n-1}x_2(t) - a_nx_1(t) + u(t) \tag{15}$$

□

Proof.

With the output definition (12), one can obtain that

$$\begin{aligned} Y(s) &= b_0U(s) + (b_1 - a_1b_0)s^{n-1}Q(s) + \cdots + (b_{n-1} - a_{n-1}b_0)sQ(s) + (b_n - a_nb_0)Q(s) \\ &= b_0U(s) + (b_1 - a_1b_0)X_n(s) + \cdots + (b_{n-1} - a_{n-1}b_0)X_2(s) + (b_n - a_nb_0)X_1(s) \end{aligned}$$

or

$$y(t) = b_0u(t) + (b_1 - a_1b_0)x_n(t) + \cdots + (b_{n-1} - a_{n-1}b_0)x_2(t) + (b_n - a_nb_0)x_1(t) \tag{16}$$

With (14), (15), and (16), one can obtain the matrices A, B, C and D of state space model as in (10). The transformation can be expressed as in following Fig.3 .

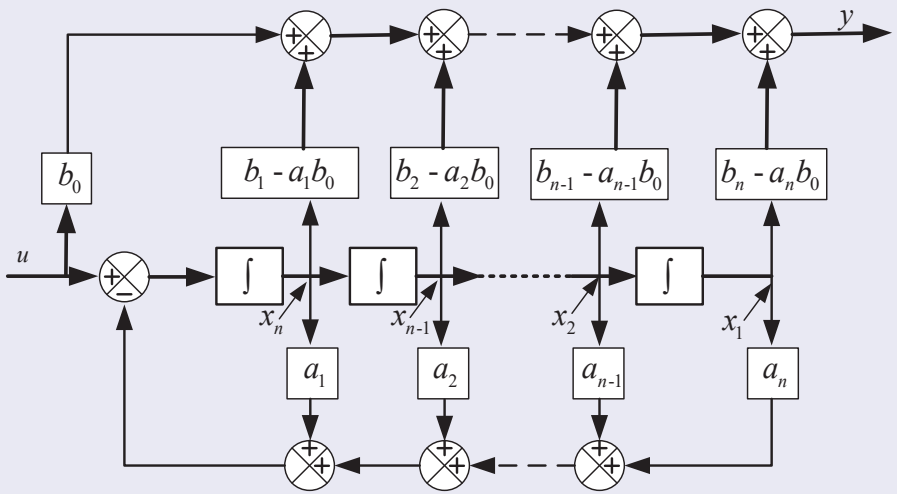


Figure 3: Block Diagram: Controllable State-space Representation

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Represent the system in (8) and (9) into observable state-space

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & -a_n \\ 1 & 0 & 1 & \dots & 0 & -a_{n-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & -a_2 \\ 0 & 0 & 0 & \dots & 1 & -a_1 \end{bmatrix}, B = \begin{bmatrix} b_n - a_n b_0 \\ b_{n-1} - a_{n-1} b_0 \\ \vdots \\ b_1 - a_1 b0 \end{bmatrix}, \tag{17}$$
$$C = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 1 \end{bmatrix}, D = b_0.$$

Proof.

Modify the transfer function (9) into the following form

$$\begin{aligned} s^n[Y(s) - b_0 U(s)] + s^{n-1}[a_1 Y(s) - b_1 U(s)] + \cdots + \\ s[a_{n-1} Y(s) - b_{n-1} U(s)] + a_n Y(s) - b_n U(s) = 0 \end{aligned} \tag{18}$$

based on (18), it can be obtained that

$$Y(s) = b_0 U(s) + \frac{1}{s}[b_1 U(s) - a_1 Y(s)] + \cdots \frac{1}{s^{n-1}}[b_{n-1} U(s) - a_{n-1} Y(s)] + \frac{1}{s^n}[b_n U(s) - a_n Y(s)] \tag{19}$$

□

Proof.

One defines the following state variables

$$\begin{aligned} X_n(s) &= \frac{1}{s}[b_1U(s) - a_1Y(s) + X_{n-1}(s)] \\ X_{n-1}(s) &= \frac{1}{s}[b_2U(s) - a_2Y(s) + X_{n-2}(s)] \\ &\vdots \\ X_2(s) &= \frac{1}{s}[b_{n-1}U(s) - a_{n-1}Y(s) + X_1(s)] \\ X_1(s) &= \frac{1}{s}[b_nU(s) - a_nY(s)] \end{aligned} \tag{20}$$

Expressing (19)

$$Y(s) = b_0U(s) + X_n(s) \tag{21}$$

□

Proof.

and substituting it into (20), it obtains that

$$\begin{aligned} sX_n(s) &= (b_1 - a_1 b_0)U(s) - a_1 X_n(s) + X_{n-1}(s) \\ sX_{n-1}(s) &= (b_2 - a_2 b_0)U(s) - a_2 X_n(s) + X_{n-2}(s) \\ &\vdots \\ sX_2(s) &= (b_{n-1} - a_{n-1} b_0)U(s) - a_{n-1} X_n(s) + X_1(s) \\ sX_1(s) &= (b_n - a_n b_0)U(s) - a_n X_n(s) \end{aligned} \tag{22}$$

And by the inverse Laplace transform, (21) can be

$$y(t) = x_n + b_0 u \tag{23}$$

and (22) can be

$$\begin{aligned} x_n(t) &= (b_1 - a_1 b_0)u(t) - a_1 x_n(t) + x_{n-1}(t) \\ x_{n-1}(t) &= (b_2 - a_2 b_0)u(t) - a_2 x_n(t) + x_{n-2}(t) \\ &\vdots \\ x_2(t) &= (b_{n-1} - a_{n-1} b_0)u(t) - a_{n-1} x_n(t) + x_1(t) \\ x_1(t) &= (b_n - a_n b_0)u(t) - a_n x_n(t) \end{aligned} \tag{24}$$

□

Proof.

Here, the matrices A, B, C and D in (17) can be verified. And the transformation can be depicted as in following Fig.4.

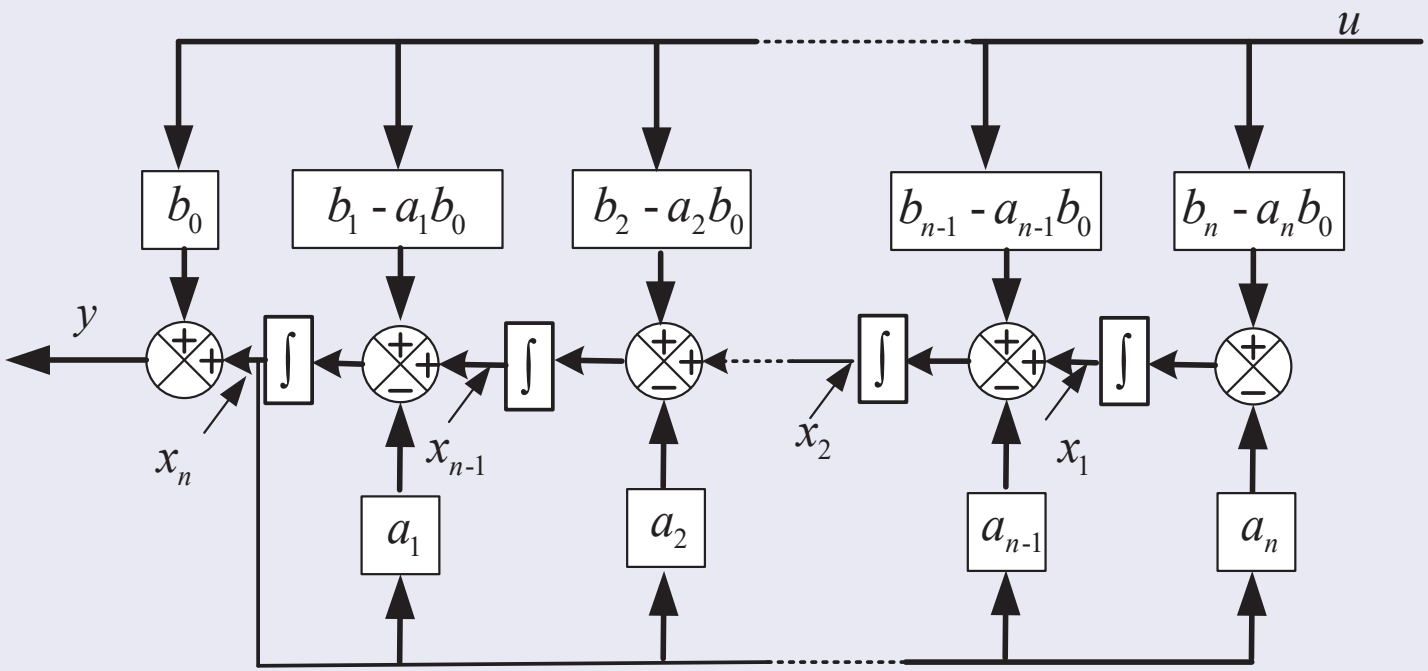


Figure 4: Block Diagram: Observable State-space Representation

□

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Represent the system in (8) into multiple distinct roots form as

$$\begin{aligned}\frac{Y(s)}{U(s)} &= \frac{b_0 s^{(n)} + b_1 s^{(n-1)} + \ldots + b_{n-1} s^{(1)} + b_n}{(s+p_1)(s+p_2)\cdots(s+p_n)} \\ &= b_0 + \frac{c_1}{s+p_1} + \frac{c_2}{s+p_2} + \ldots + \frac{c_n}{s+p_n}\end{aligned} \tag{25}$$

The diagonal canonical form state-space representation can be given as

$$\begin{aligned}A &= \begin{bmatrix} -p_1 & 0 & \ldots & 0 \\ 0 & -p_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & -p_1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \\ C &= \begin{bmatrix} c_1 & c_2 & \ldots & c_n \end{bmatrix}, D = b_0.\end{aligned} \tag{26}$$

Proof.

With the expression in (25), one can obtain that

$$Y(s) = b_0 U(s) + \frac{c_1}{s+p_1}U(s) + \frac{c_2}{s+p_2}U(s) + \ldots + \frac{c_n}{s+p_n}U(s) \tag{27}$$

□

Proof.

Define the variables

$$\begin{cases} X_1(s) &= \dfrac{1}{s+p_1}U(s) \\ X_2(s) &= \dfrac{1}{s+p_2}U(s) \\ &\vdots \\ X_n(s) &= \dfrac{1}{s+p_n}U(s) \end{cases} \tag{28}$$

It can be obtained that

$$\begin{cases} sX_1(s) &= -p_1X_1(s)+U(s) \\ sX_2(s) &= -p_2X_2(s)+U(s) \\ &\vdots \\ sX_n(s) &= -p_nX_n(s)+U(s) \end{cases} \tag{29}$$

and (27)

$$Y(s) = b_0U(s) + c_1X_1(s) + c_2X_2(s) + \ldots + c_nX_n(s) \tag{30}$$

□



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Proof.

Via the inverse Laplace transformation, it can be given that

$$\begin{cases} \dot{x}_1 &= -p_1 x_1(t) + u \\ \dot{x}_2 &= -p_2 x_2(t) + u \\ &\vdots \\ \dot{x}_n &= -p_n x_n(t) + u \end{cases} \tag{31}$$

and

$$y(s) = b_0 u(t) + c_1 x_1(t) + c_2 x_2(t) + \ldots + c_n x_n(t) \tag{32}$$

Such that the matrices A, B, C and D in (26) can be obtained. The transformation can be depicted as in following Fig.4. □



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Proof.

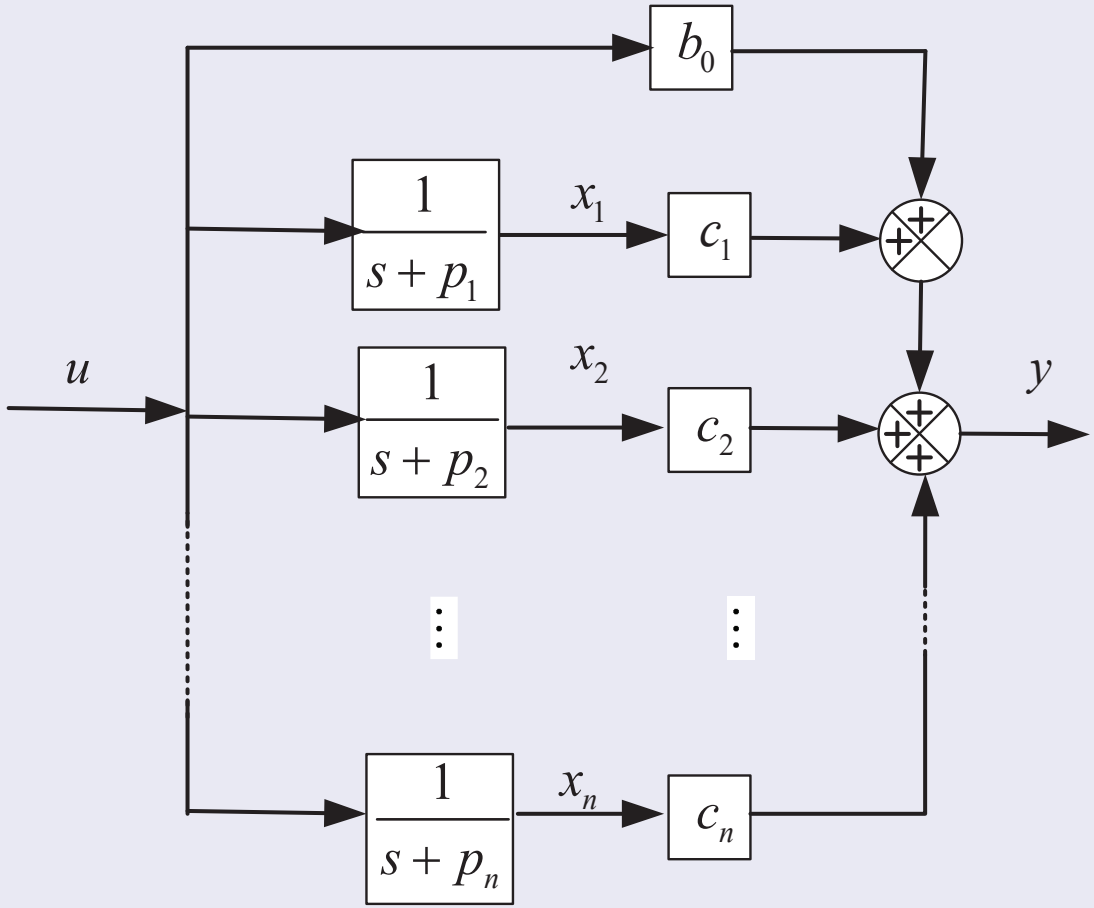


Figure 5: Block Diagram: Diagonal State-space Representation

□

Alternative Representation

If one chooses

$$\begin{aligned} X_1(s) &= \frac{c_1}{s+p_1}U(s) \\ X_2(s) &= \frac{c_2}{s+p_2}U(s) \\ &\vdots \\ X_n(s) &= \frac{c_n}{s+p_n}U(s) \end{aligned} \tag{33}$$

it has that

$$\begin{aligned} sX_1(s) &= -p_1X_1(s) + c_1U(s) \\ sX_2(s) &= -p_2X_2(s) + c_2U(s) \\ &\vdots \\ sX_n(s) &= -p_nX_n(s) + c_nU(s) \end{aligned} \tag{34}$$

and

$$Y(s) = b_0U(s) + X_1(s) + X_2(s) + \cdots + X_n(s) \tag{35}$$

By the inverse Laplace transformation, it has that

$$\begin{cases} \dot{x}(t) &= Ax(t) + C^Tu \\ y &= B^Tx(t) + Du \end{cases} \tag{36}$$

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Represent the system (8) into multiple distinct roots form as

$$\begin{aligned}\frac{Y(s)}{U(s)} &= \frac{b_0 s^{(n)} + b_1 s^{(n-1)} + \ldots + b_{n-1} s^{(1)} + b_n}{(s+p_1)^3 (s+p_4)(s+p_5)\cdots(s+p_n)} \\ &= b_0 + \frac{c_1}{(s+p_1)^3} + \frac{c_2}{(s+p_1)^2} + \frac{c_3}{(s+p_1)} + \frac{c_4}{(s+p_4)} + \ldots + \frac{c_n}{s+p_n}\end{aligned} \tag{37}$$

The state-space representation of the Jordan form is given by

$$A = \begin{bmatrix} -p_1 & 1 & 0 & 0 & \ldots & 0 \\ 0 & -p_1 & 1 & 0 & \ldots & 0 \\ 0 & 0 & -p_1 & 0 & \ldots & 0 \\ 0 & \ldots & 0 & -p_4 & \ldots & 0 \\ \vdots & \ddots & \vdots & 0 & \ddots & 0 \\ 0 & \cdot & 0 & 0 & 0 & -p_n \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \tag{38}$$

$$C = \begin{bmatrix} c_1 & c_2 & \ldots & c_n \end{bmatrix}, D = b_0.$$



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Proof.

With the equation (37), one can obtain that

$$\begin{aligned} Y(s) &= b_0U(s) + \frac{c_1}{(s+p_1)^3}U(s) + \frac{c_2}{(s+p_1)^2}U(s) + \frac{c_3}{(s+p_1)}U(s) \\ &\quad + \frac{c_4}{(s+p_4)}U(s) + \ldots + \frac{c_n}{s+p_n}U(s) \end{aligned} \tag{39}$$

Define that

$$\begin{aligned} X_1(s) &= \frac{1}{(s+p_1)^3}U(s) \\ X_2(s) &= \frac{1}{(s+p_1)^2}U(s) \\ X_3(s) &= \frac{1}{(s+p_1)}U(s) \\ &\vdots \\ X_n(s) &= \frac{1}{s+p_n}U(s) \end{aligned} \tag{40}$$

□

Proof.

Notice

$$\begin{aligned} \frac{X_1(s)}{X_2(s)} &= \frac{1}{s+p_1} \\ \frac{X_2(s)}{X_3(s)} &= \frac{1}{s+p_1} \end{aligned} \tag{41}$$

One further obtains that

$$\begin{cases} sX_1(s) &= -p_1X_1(s) + X_2(s) \\ sX_2(s) &= -p_1X_1(s) + X_3(s) \\ sX_3(s) &= -p_1X_3(s) + U(s) \\ sX_4(s) &= -p_4X_4(s) + U(s) \\ &\vdots \\ sX_n(s) &= -p_nX_n(s) + U(s) \end{cases} \tag{42}$$

and

$$Y(s) = b_0U(s) + c_1X_1(s) + c_2X_2(s) + c_3X_3(s) + c_4X_4(s) + \cdots + c_nX_n(s) \tag{43}$$

□



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Proof.

By inverse Laplace transformation, the system state and output in time domain can be verified. Parameter matrices A, B, C and D are in (38). The transformation can be depicted as in following Fig.6.

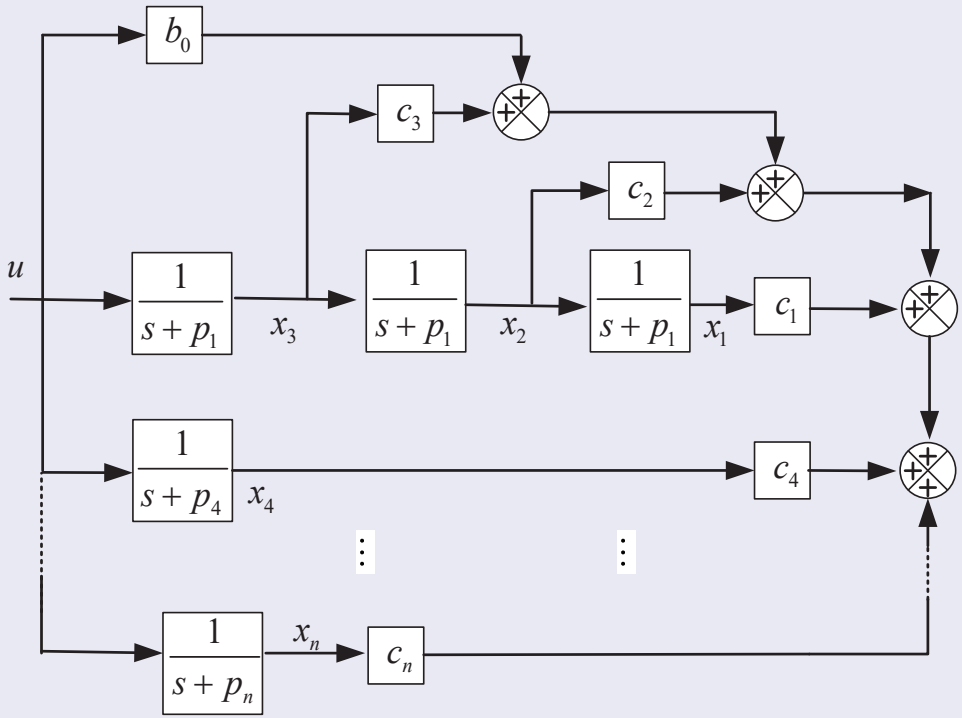


Figure 6: Block Diagram: Jordan State-space Representation

□

3 State - space Representation of Transfer - function Systems

- Controllable State-space Representation
- Observable State-space Representation
- Diagonal Canonical State-space Representation
- Jordan Canonical State-space Representation
- From State-space Model to Laplace Model
- Reformation of State-space Model into Diagonal or Jordan Forms



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The Laplace presentation can be obtained via

$$G(s) = C(sI - A)^{-1}B + D \tag{44}$$

Remark

For MIMO system $C = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_p \end{bmatrix} \in \mathcal{R}^{p\times n}$, $D = \begin{bmatrix} D_{11} & D_{12} & \cdots & D_{1m} \\ D_{21} & D_{22} & \cdots & D_{2m} \\ \vdots & \vdots & \cdots & \vdots \\ D_{p1} & D_{p2} & \cdots & D_{pm} \end{bmatrix} \in \mathcal{R}^{p\times m}$,

$B = \begin{bmatrix} B_1 & B_2 & \cdots & B_m \end{bmatrix} \in \mathcal{R}^{n\times m}$, *how to obtain the transfer functions?*

Example

Consider the system in Laplace form below

$$\frac{Y(s)}{U(s)} = \frac{s+3}{s^2+3s+2}$$

- Represent the transformation into observable, controllable, and diagonal forms.
- Represent the state-space form system into the Laplace form, and check whether the results are the same.

3 State - space Representation of Transfer - function Systems

- Controllable State-space Representation
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Definition (Eigenvalues of A:)

Eigenvalues of A, also named characteristic roots, are the roots of the characteristic equation

$$|\lambda I - A| = 0 \tag{45}$$

Example

Calculate the eigenvalues of A

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$$

Diagonalization: If A is with distinct eigenvalues and

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \dots & -a_2 & -a_1 \end{bmatrix}$$

the transformation $x = Pz$



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$$P = \begin{bmatrix} 1 & 1 & \dots & 1 \\ \lambda_1 & \lambda_2 & \dots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \dots & \lambda_n^2 \\ \vdots & \vdots & \vdots & \vdots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \dots & \lambda_n^{n-1} \end{bmatrix}$$

It has that

$$P^{-1}AP = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

For system with multiple eigenvalues, diagonalization is impossible. For example, if $A \in \mathcal{R}^{3\times 3}$,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix}$$

has the eigenvalues that λ_1, λ_1, and λ_3, the transformation would be $x = Sz$, where

$$S = \begin{bmatrix} 1 & 0 & 1 \\ \lambda_1 & 1 & \lambda_3 \\ \lambda_1^2 & \lambda_1 & \lambda_3^2 \end{bmatrix}$$



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It yields that

$$S^{-1}AS = \begin{bmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

Invariance of Eigenvalues: Via substituting the system state $z = P^{-1}x$ or $z = S^{-1}x$, it obtains that reformed system state

$$\begin{cases} \dot{z} &= A_z z(t) + B_z u(t) \\ y &= C_z z(t) + D_z u(t) \end{cases} \tag{46}$$

Theorem

It would find that the transformed state-space model is with the same eigenvalues as in the original system, which means that the system transformation doesn't change the stability result.

Proof.

$$\begin{aligned} |\lambda I - P^{-1}AP| &= |\lambda P^{-1}P - P^{-1}AP| \\ &= |P^{-1}(\lambda I - A)P| \\ &= |P^{-1}||\lambda I - A||P| \\ &= |P^{-1}||P||\lambda I - A| \\ &= |\lambda I - A| \end{aligned} \tag{47}$$

□

Non-uniqueness of State Variables and State-space Model

The reformed state variable $z = P^{-1}x$, here it offers a alternative typical transformation with the nonsingular matrix P, but this is not the unique transformation. Alternative transformation can be offered via the new definition P. The state-space models of the system are not unique.

Example

Consider the following state space model below

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} u \tag{48}$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x$$

Transform the given model into diagonal or Jordan standard form.



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4 Solving the Time-invariant State Equation

- Solution of Homogeneous State Equations
  - Solution by the Matrix Series
  - Solution by the Laplace Transformation
  - State-Transition Matrix
- Solution of Nonhomogeneous State Equations
  - Solution by State-transition Matrix
  - Laplace Transform Approach
  - Solution from Initial t_0

For the homogeneous system

$$\dot{x}(t) = Ax(t) \tag{49}$$

where $A \in \mathcal{R}^{n\times n}$. Via the series expansion theory around t, it can be assumed that the solution can be in the form that

$$x(t) = \mathbf{b}_0 + \mathbf{b}_1 t .... = \sum_{i=0}^{\infty} \mathbf{b}_i t^i \tag{50}$$

where $\mathbf{b}_i \in \mathcal{R}^n$, if one sets $t = 0$ in (50), it obtains $b_0 = x(0)$. Substituting (50) into left and right sides of (49), one obtains that

$$\sum_{i=1}^{\infty} \left((i)\mathbf{b}_i t^{i-1} \right) = A \sum_{i=1}^{\infty} \mathbf{b}_{i-1} t^{i-1}$$

And, it further verifies that

$$\begin{aligned}
\mathbf{b}_0 &= x(0) \\
\mathbf{b}_1 &= A\mathbf{b}_0 \\
\mathbf{b}_2 &= \frac{1}{2!}A\mathbf{b}_1 = \frac{1}{2!}A^2\mathbf{b}_0 \\
\mathbf{b}_3 &= \frac{1}{3!}A\mathbf{b}_2 = \frac{1}{3!}A^3\mathbf{b}_0 \\
&\vdots \\
\mathbf{b}_k &= \frac{1}{k!}A^k\mathbf{b}_0
\end{aligned} \tag{51}$$



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Definition

Matrix Exponential: the matrix exponential of $A \in \mathcal{R}^{n\times n}$, converges absolutely for any time t.

$$\sum_{i=0}^{\infty} \frac{A^i t^i}{i!} = e^{At} \tag{52}$$

Therefore, it obtains that

$$x(t) = \left(\sum_{i=0}^{\infty} \frac{A^i t^i}{i!}\right) x(0) = e^{At} x(0) \tag{53}$$

Lemma

The matrix exponential has the property that

$$\begin{aligned} e^{At}e^{As} &= \left(\sum_{i=0}^{\infty} \frac{A^i t^i}{i!}\right)\left(\sum_{i=0}^{\infty} \frac{A^i s^i}{i!}\right) = \sum_{k=0}^{\infty} A^k \left(\sum_{i=0}^{\infty} \frac{t^i s^{k-1}}{i!(k-i)!}\right) \\ &= \sum_{k=0}^{\infty} A^k \frac{(t+s)^k}{k!} = e^{A(t+s)} \end{aligned} \tag{54}$$

If $s = -t$, it has that $e^{At}e^{-At} = e^{A(t-t)}$

Lemma

$$\begin{array}{rcl} e^{(A+B)t} & = & e^{At}e^{Bt}, \text{ if } AB = BA \\ e^{(A+B)t} & \neq & e^{At}e^{Bt}, \text{ if } AB \neq BA \end{array} \tag{55}$$

Proof.

According to Definition.6, it has that

$$\begin{array}{rcl} e^{(A+B)t} & = & \sum_{i=0}^{\infty} \dfrac{(A+B)^i t^i}{i!} \\ e^{At}e^{Bt} & = & \left(\sum_{i=0}^{\infty} \dfrac{A^i t^i}{i!}\right)\left(\sum_{i=0}^{\infty} \dfrac{B^i t^i}{i!}\right) \end{array}$$

It further has

$$\begin{aligned} e^{(A+B)t} - e^{At}e^{Bt} &= \frac{(BA-AB)t^2}{2!} \\ &+ \frac{(BA^2 + ABA + B^2A + BAB - 2A^2B - 2AB^2)t^3}{3!} + f(A,B)(AB-BA) \end{aligned} \tag{56}$$

that if $AB - BA = 0$, (56)=0, and the Lemma holds. Either it cannot holds. □

4 Solving the Time-invariant State Equation
- Solution of Homogeneous State Equations
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According to the laplace transformation, for (49), it has

$$sX(s) - x(0) = AX(s) \tag{57}$$

therefore, one obtains that

$$(sI - A)X(s) = x(0)$$

and

$$X(s) = (sI - A)^{-1}x(0)$$

And, by the inverse Laplace Transformation, it has

$$x(t) = \mathfrak{L}^{-1}(X(s)) = \mathfrak{L}^{-1}\left(sI - A\right)^{-1}x(0) \tag{58}$$

For that

$$(sI - A)^{-1} = \sum_{i=0}^{\infty} \frac{A^i}{s^{i+1}}$$

one obtains

$$\mathfrak{L}^{-1}\left[(sI - A)^{-1}\right] = \sum_{i=0}^{\infty} \frac{A^i t^i}{i!}$$

And, according to Definition.6, one has that

$$\mathfrak{L}^{-1}\left[(sI - A)^{-1}\right] = e^{At}$$

Remark

Mathematics bridges the two solutions, Series and Laplace Transform Approach, together.

4 Solving the Time-invariant State Equation
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Solution for (49) can be

$$x(t) = \Phi(t)x(0) \tag{59}$$

where $\Phi(t) \times \mathcal{R}^{n\times n}$, is the unique solution of

$$\dot{\Phi}(t) = A\Phi(t) \tag{60}$$

and $\Phi(0) = I$. Note that

$$x(0) = \Phi(0)x(0)$$

and

$$\dot{x}(t) = Ax(t) = \dot{\Phi}(t)x(0) = A\Phi(t)x(0)$$

one can further verify that (59) holds. According to (53), and (58), one can further define that

$$\Phi(t) = e^{At} = \mathfrak{L}^{-1}\left[(sI - A)^{-1}\right] \tag{61}$$

Properties of State-Transition Marices:

For the time-invariant system

$$\dot{x} = Ax$$

the State-Transition matrix is with the property that

- $\Phi(0) = e^{A0} = \mathrm{I}$, $[\Phi(t)]^n = \Phi(nt)$
- $\Phi(t) = e^{At} = (e^{-At})^{-1} = (\Phi(-t))^{-1}$ or $\Phi(-t) = \Phi^{-1}(t)$
- $\Phi(t_1 + t_2) = e^{A(t_1+t_2)} = e^{At_1}e^{At_2} = \Phi(t_1)\Phi(t_2) = \Phi(t_2)\Phi(t_1)$
- $\Phi(t_2 - t_1)\Phi(t_1 - t_0) = \Phi(t_2 - t_0) = \Phi(t_1 - t_0)\Phi(t_2 - t_1)$

Example

Obtain the state-transition matrix $\Phi(t)$ of the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{aligned} \left[(sI - A)^{-1}\right] &= \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \\ &= \begin{bmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{-2}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix} \end{aligned}$$

The state-transition matrix $\Phi(t)$ is given by

$$\begin{aligned} \Phi(t) &= e^{At} = \mathfrak{L}^{-1}\left[(sI - A)^{-1}\right] \\ &= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \end{aligned}$$

Equality

'left plane root', 'minus real part poles', 'Routh-Hurwitz criteria', 'eigenvalues of A', 'e^{At}'

4 Solving the Time-invariant State Equation

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Consider the nonhomogeneous state equation

$$\dot{x} = Ax + Bu \rightarrow \dot{x} - Ax = Bu \tag{62}$$

where $x \in \Re^n$, $u \in \Re^m$, $A \in \Re^{n\times n}$, $B \in \Re^{n\times m}$. Left multiply e^{-At} both sides, it obtains

$$e^{-At}[\dot{x} - Ax] = \frac{d(e^{-At}x(t))}{dt} = e^{-At}Bu \tag{63}$$

Integrate the second and third terms in the interval $[0, t]$

$$e^{-At}x(t) - x(0) = \int_0^t e^{-A\tau}Bu(\tau)d\tau \tag{64}$$

left multiply e^{At} both sides, it has

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \tag{65}$$

or

$$x(t) = \underbrace{\int_0^t \Phi(t-\tau)Bu(\tau)d\tau}_{\text{Input transition}} + \underbrace{\Phi(t)x(0)}_{\text{Initial state transition}} \tag{66}$$

Φ is the State-transition matrix defined above.

4 Solving the Time-invariant State Equation

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By Laplace transform approach, (62) can be expressed as

$$sX(s) - x(0) = AX(s) + BU(s) \tag{67}$$

It further yields that

$$(sI - A)X(s) = x(0) + BU(s)$$

and

$$X(s) = (sI - A)^{-1}(x(0) + BU(s))$$

With (61), $\Phi(t) = \mathfrak{L}^{-1}\left[(sI - A)^{-1}\right] = e^{At}$, it has

$$X(s) = \mathfrak{L}[e^{At}](x(0) + BU(s))$$

and, by inverse Laplace transform it verifies that

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \tag{68}$$



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4 Solving the Time-invariant State Equation
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The solution based on Laplace transform in (68) is under the zero time, i.e. initialization time is 0. When the initial time is t_0 rather than zero, the solution is

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)d\tau \tag{69}$$

Remark

Not only the solution for the state-space model, but it also offers a method on solving the iteration for the continuous system model.

Example

Obtain the solution of the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

where u is the unit-step function from $t = 0$. According to (68), one can have that

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} + \int_0^t \begin{bmatrix} 2e^{-(t-\tau)} - e^{-2(t-\tau)} & e^{-(t-\tau)} - e^{-2(t-\tau)} \\ -2e^{-(t-\tau)} + 2e^{-2(t-\tau)} & -e^{-(t-\tau)} + 2e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} 1 d\tau$$

Example

$$
\begin{aligned}
\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} &= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} \\
&\quad + \int_0^t \begin{bmatrix} e^{-(t-\tau)} - e^{-2(t-\tau)} \\ -e^{-(t-\tau)} + 2e^{-2(t-\tau)} \end{bmatrix} d\tau \\
&= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} \\
&\quad + \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}
\end{aligned}
$$



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1 Introduction

2 State-space Equation Model

3 State - space Representation of Transfer - function Systems

4 Solving the Time-invariant State Equation

5 Controllability and Observability of System

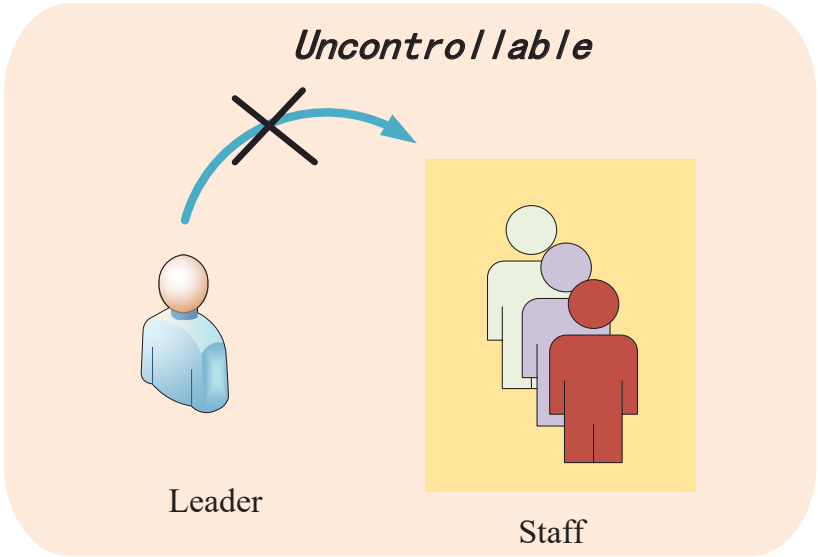
6 Pole Placement

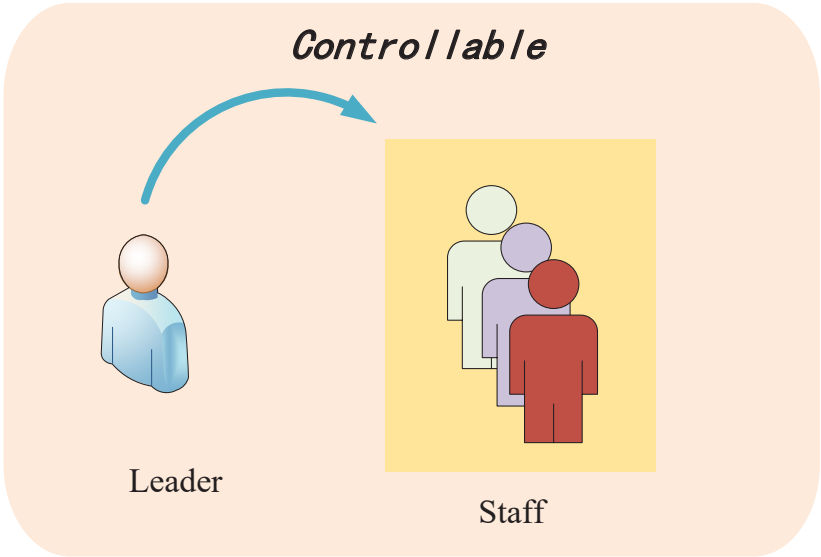
7 Stability Analysis based on Lyapunov Criterion

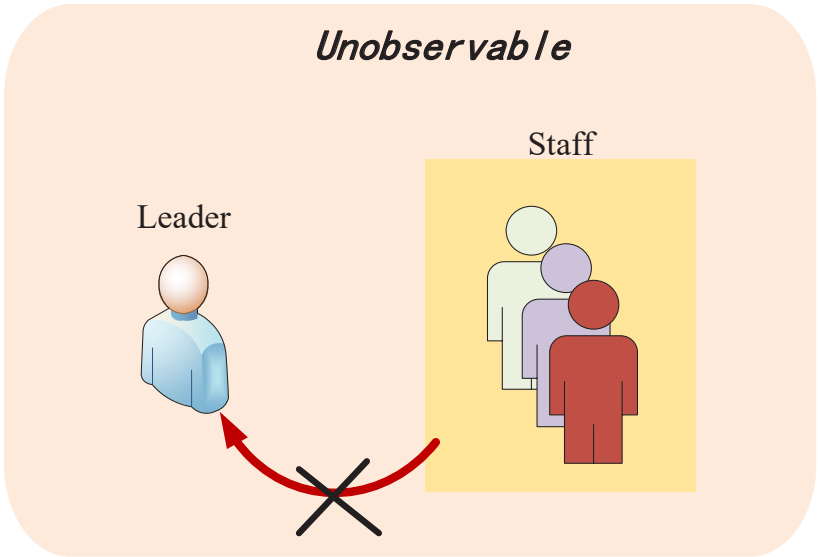
8 Controller based on Observer

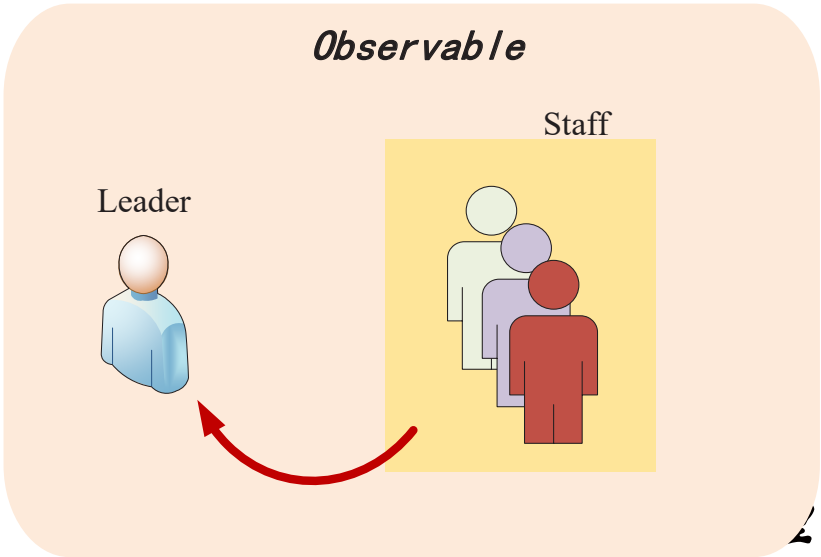
9 Chapter Conclusion

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Uncontrollable
Leader
Staff


Controllable
Leader
Staff


Unobservable
Staff
Leader


Observable
Staff
Leader

Definition (Controllability)

A system is said to be **controllable** at time t_0, if it is possible utilizing an unconstrained control vector to transfer the system from any initial state $x(t_0)$ to any other state in a finite interval of time.

Definition (Observability)

A system is said to be **observable** at time t_0, with the system in state $x(t_0)$, if it is possible to determine this state from the observation of the output over a finite time interval.



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5 Controllability and Observability of System
- Complete Controllability of Continuous-Time Systems
- Complete Observability of Continuous-Time Systems
- Principle of Duality



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Consider the following system

$$\dot{x} = Ax + Bu \tag{70}$$

where $x \in \mathcal{R}^n$, $u \in \mathcal{R}^1$, $A \in \mathcal{R}^{n\times n}$, $B \in \mathcal{R}^{n\times 1}$.

Definition

Equation (70) is **state controllable** at $t = t_0$, if it is possible to construct an unconstraint control signal that will transfer an initial state to any final state in a finite time interval $t_0 \leqslant t \leqslant t_1$.

Definition

If every state is controllable, then the system is said to be **completely state controllable**.

The solution of (70) is

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \tag{71}$$

Assume the final state at t_1 is the origin of the state-space, according to the definition of complete state controllability, one has

$$x(t_1) = 0 = e^{At}x(0) + \int_0^{t_1} e^{A(t_1-\tau)}Bu(\tau)d\tau$$

and further obtains

$$\begin{aligned} x(0) &= -e^{-At}\int_0^{t_1} e^{A(t_1-\tau)}Bu(\tau)d\tau \\ &= -\int_0^{t_1} e^{-A\tau}Bu(\tau)d\tau \end{aligned} \tag{72}$$



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According to the series expanding, the expression $e^{-A\tau}$ can be written as

$$e^{-A\tau} = \sum_{i=0}^{n-1} \frac{A^i \tau^i}{i!} = \sum_{i=0}^{n-1} \alpha_i(\tau) A^i \tag{73}$$

(72) can be further expressed as

$$\begin{aligned} x(0) &= -\int_0^{t_1} e^{-A\tau} B u(\tau) d\tau \\ &= -\int_0^{t_1} \left(\sum_{i=0}^{n-1} \alpha_i(\tau) A^i B u(\tau) \right) d\tau \\ &= -\sum_{i=0}^{n-1} A^i B \beta_i \\ &= -\left[B | AB | A^2 B | \cdots | A^{n-1} B \right] \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{bmatrix} \end{aligned} \tag{74}$$

$\beta_i = \int_0^{t_1} \alpha_k(\tau) u(\tau) d\tau$ is unnecessary to be known. If the system is completely controllable, then for any given initial state $x(0)$, (74) holds, which means that the rank of the matrix below is n, i.e.

$$Rank([B|AB|A^2B|\cdots|A^{n-1}B]) = n$$



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The further result can be expanded to the situation that the control input u is of r-dimension, $r > 1$. The condition for complete state controllability is whether

$$Rank([B|AB|A^2B|\cdots|A^{n-1}B]) = n$$

Definition

The matrix

$$\left[B|AB|A^2B|\cdots|A^{n-1}B\right]$$

is defined as the **controllability matrix**.

Example

Consider the below system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

analyze its controllability.



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5 Controllability and Observability of System
- Complete Controllability of Continuous-Time Systems
- Complete Observability of Continuous-Time Systems
- Principle of Duality

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Consider the unforced system

$$\begin{cases} \dot{x} &= Ax \\ y &= Cx \end{cases} \tag{75}$$

where $x \in \mathcal{R}^n$, $y \in \mathcal{R}^p$, $A \in \mathcal{R}^{n\times n}$, and $C \in \mathcal{R}^{p\times n}$.

Definition

The system is said to be **completely observable** if every state $x(t_0)$ can be determined from the observation of $y(t)$ over a finite time interval $t_0 \le t \le t_1$.

If the system is described as following

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{cases} \tag{76}$$

then the solutions of $x(t)$ and $y(t)$ are

$$\begin{cases} x(t) &= e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \\ y(t) &= Cx(t) + Du = Ce^{At}x(0) + C\int_0^t e^{A(t-\tau)}Bu(\tau)d\tau + Du(t) \end{cases} \tag{77}$$



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Complete Observability of Continuous-time Systems：
One can verify the output $y(t)$

$$\begin{aligned} y(t) &= Ce^{At}x(0) = \sum_{i=0}^{n-1} \alpha_k(t) CA^k x(0) \\ &= \begin{bmatrix} \alpha_0 & \alpha_1 & \cdots & \alpha_{n-1} \end{bmatrix} \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} x_0 \end{aligned} \tag{78}$$

Parameter α_i for $i = 0, \cdots, n-1$ is neither required. If the system is completely observable, $x(0)$ is uniquely determined from (78), which means that it requires the rank of

$$Rank(\Psi) = n \tag{79}$$

or Ψ has n linearly independent column vectors, where

$$\Psi = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} \in \Re^{(p*n)\times n}$$



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Example

Consider the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

controllable and observable?

5 Controllability and Observability of System

- Complete Controllability of Continuous-Time Systems
- Complete Observability of Continuous-Time Systems
- Principle of Duality

Consider two systems $\mathfrak{S}_1$ and $\mathfrak{S}_2$ below

$$\mathfrak{S}_1 \begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx \end{cases} \tag{80}$$

$$\mathfrak{S}_2 \begin{cases} \dot{z} &= A^T z + C^T v \\ n &= B^T z \end{cases} \tag{81}$$

Definition (Principle of Duality)

The system $\mathfrak{S}_1$ is completely state controllable (observable) if and only if system $\mathfrak{S}_2$ is completely observable (state controllable).

The definition indicates that the observability of a given system can be checked by testing the state controllability of its dual.

Definition

For a partially observable system, if the unobservable modes are stable and the observable modes are unstable, the system is said to be **detectable**.

Remark

*The concept of **detectability** is dual to the concept of **stabilizability** for a partially controllable system*

6 Pole Placement

- Sufficient and Necessary Condition of Pole-placement
- Methods on Designing K



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Consider an SISO state-space model below

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{cases} \tag{82}$$

a state-feedback controller is designed that

$$u = -Kx \tag{83}$$

- It covers the PID controller;
- $K \in \mathcal{R}^{m \times n}$;
- The controller can be $u = Kx$.

The closed-loop system is

$$\dot{x}(t) = (A - BK)\, x(t) \tag{84}$$

and the solution of this equation is

$$x(t) = e^{(A-BK)t} x(0) \tag{85}$$



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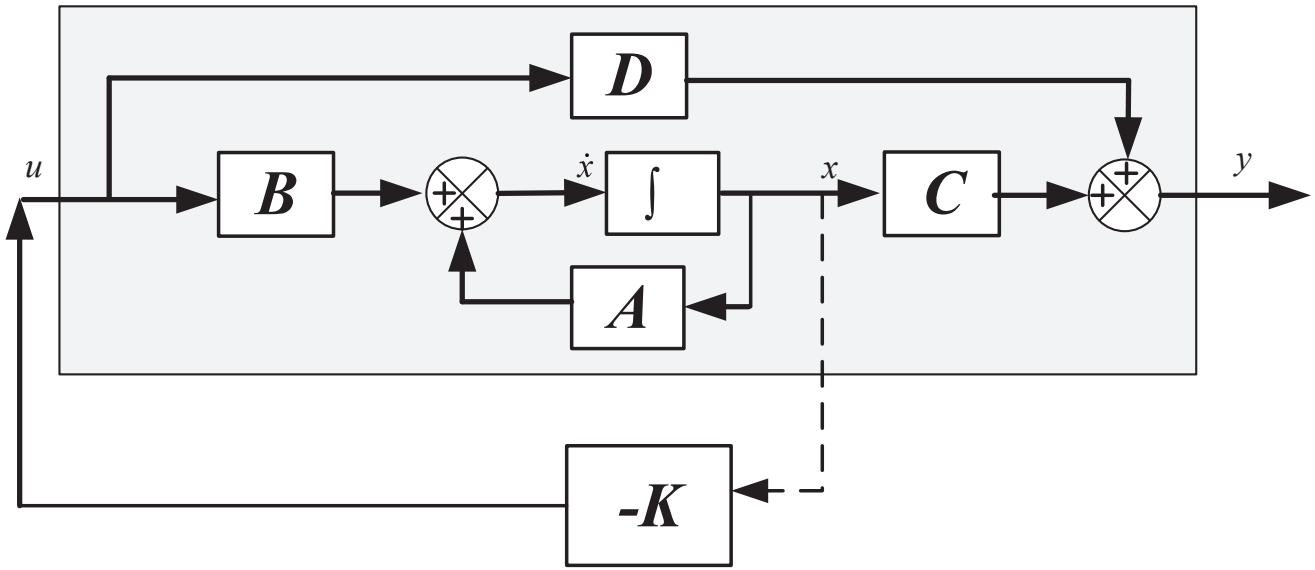


Figure 7: Feedback Controller under State-space

Lemma

The necessary and sufficient condition for arbitrary pole placement is that the system is ***completely state controllable****.*

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Proof.

Necessary condition analysis:

Suppose (82) is not completely state controllable,

$$Rank\left[\begin{array}{cccc} B & AB & \cdots & A^{n-1}B \end{array}\right] = q < n$$

and there are q linearly independent column vectors, $f_1, f_2, \cdots, f_q$, in the controllability matrix. And, we select $n-q$ additional vectors $v_{q+1}, v_{q+2}, \cdots, v_n$ that

$$P = \left[\begin{array}{ccccccc} f_1 & f_2 & \cdots & f_q & v_{q+1} & v_{q+2} & \cdots & v_n \end{array}\right]$$

is of rank n. Then it can be verified that

$$\hat{A} = P^{-1}AP = \left[\begin{array}{cc} A_{11} & A_{12} \\ 0 & A_{22} \end{array}\right], \hat{B} = P^{-1}B = \left[\begin{array}{c} B_{11} \\ 0 \end{array}\right]$$

Define

$$\hat{K} = KP = \left[\begin{array}{cc} k_1 & k_2 \end{array}\right]$$

□

Proof.

one has that

$$
\begin{aligned}
|sI - A + BK| &= \left|P^{-1}(sI - A + BK)P\right| \\
&= \left|sI - P^{-1}AP + P^{-1}BKP)\right| \\
&= \left|sI - \hat{A} + \hat{B}\hat{K})\right| \\
&= \left|sI - \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} + \begin{bmatrix} B_{11} \\ 0 \end{bmatrix} \begin{bmatrix} k_1 & k_2 \end{bmatrix}\right| \\
&= \begin{vmatrix} sI - A_{11} + B_{11}k_1 & -A_{12} + B_{11}k_2 \\ 0 & sI - A_{22} \end{vmatrix} \\
&= |sI - A_{11} + B_{11}k_1|\,|sI - A_{22}|
\end{aligned}
$$

- The eigenvalues of A_{22} do not depend on the K;
- If the system is not completely state controllable, then eigenvalues of matrix A that cannot be arbitrarily placed.
- To place the eigenvalues of matrix $A - BK$ arbitrarily, the system should be completely state controllable.

Proof.

Sufficient condition analysis: if the system is completely state controllable, then all eigenvalues of matrix A can be arbitrarily placed.
Define a transformation matrix T that

$$T = MW \tag{86}$$

where M is the controllable matrix

$$M = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} \tag{87}$$

and

$$W = \begin{bmatrix} a_{n-1} & a_{n-2} & \cdots & a_1 & 1 \\ a_{n-2} & a_{n-3} & \cdots & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ a_1 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

□

Proof.

a_i is the coefficients of the characteristic polynomial

$$|sI - A| = s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n$$

Define a new state vector $\hat{x}$ by

$$x = T\hat{x} \tag{88}$$

If the rank of the controllability matrix M is n, the inverse of matrix T exists, such that (82) can be modified into

$$\dot{\hat{x}} = T^{-1}AT\hat{x} + T^{-1}Bu \tag{89}$$

where

$$T^{-1}AT = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}, T^{-1}B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \tag{90}$$

□

Proof.

If the desired eigenvalues $\mu_1, \mu_2, \cdots, \mu_n$, the desired characteristic equation is

$$(s-\mu_1)(s-\mu_2)\cdots(s-\mu_n) = s^n+\alpha_1 s^{n-1}+\alpha_2 s^{n-2}+\cdots+\alpha_{n-2}s^2+\alpha_{n-1}s+\alpha_n \quad (91)$$

If $u = -KT\hat{x}$ is used to control the system (89). Reforming

$$KT = \begin{bmatrix} \delta_n & \delta_{n-1} & \cdots & \delta_1 \end{bmatrix} \quad (92)$$

thus the system becomes

$$\dot{\hat{x}} = T^{-1}AT\hat{x} - T^{-1}BKT\hat{x}$$

and the characteristic equation is

$$\begin{aligned} \left|sI - T^{-1}AT + T^{-1}BKT\right| &= 0 \\ &= \begin{vmatrix} s & -1 & \cdots & 0 \\ 0 & s & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ a_n+\delta_n & a_{n-1}+\delta_{n-1} & \cdots & s+a_1+\delta_1 \end{vmatrix} \end{aligned}$$

Proof.

With the desired eigenvalues, one gets that

$$\begin{aligned} a_1 + \delta_1 &= \alpha_1 \\ a_2 + \delta_2 &= \alpha_1 \\ &\vdots \\ a_n + \delta_n &= \alpha_n \end{aligned} \tag{93}$$

one can derive the gain K

$$\begin{aligned} K &= \begin{bmatrix} \delta_n & \delta_{n-1} & \cdots & \delta_1 \end{bmatrix} T^{-1} \\ &= \begin{bmatrix} \alpha_n - a_n & \alpha_{n-1} - a_{n-1} & \cdots & \alpha_1 - a_1 \end{bmatrix} T^{-1} \end{aligned} \tag{94}$$

□

6 Pole Placement

- Sufficient and Necessary Condition of Pole-placement
- Methods on Designing K

Using Transformation Matrix

- Step 1: Controllability condition checking. It terminates if it's not controllable.
- Step 2: The characteristic polynomial for matrix A is

$$|sI - A| = s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n$$

One obtains the values of $a_1, a_2, \cdots, a_n$.

- Step 3: Define a transformation matrix T that

$$T = MW \tag{95}$$

where M is the controllable matrix

$$M = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} \tag{96}$$

and

$$W = \begin{bmatrix} a_{n-1} & a_{n-2} & \cdots & a_1 & 1 \\ a_{n-2} & a_{n-3} & \cdots & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ a_1 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

Using Transformation Matrix

- Step 4: Using the desired eigenvalues, the desired characteristic polynomial is

$$(s-\mu_1)(s-\mu_2)\cdots(s-\mu_n) = s^n + \alpha_1 s^{n-1} + \alpha_2 s^{n-2} + \cdots + \alpha_{n-2}s^2 + \alpha_{n-1}s + \alpha_n \quad (97)$$

  determine the values of $\alpha_1, \alpha_2, \cdots, \alpha_n$.
- Step 5: Calculate the K

$$K = \begin{bmatrix} \alpha_n - a_n & \alpha_{n-1} - a_{n-1} & \cdots & \alpha_1 - a_1 \end{bmatrix} T^{-1} \quad (98)$$

Write the state-feedback gain

$$K = \begin{bmatrix} k_1 & k_2 & \cdots & k_n \end{bmatrix} \quad (99)$$

and substitute the gain matrix K into the characteristic polynomial. Equal it to the desired root characteristic polynomial equations

$$|sI - A + BK| = \prod_{i=1}^{n} (s - \mu_i) \quad (100)$$

such that the element k_i can be obtained.

Remark

This approach is convenient if $n = 1, 2, 3$, but tedious and complicated if $n \geq 4$.

Example

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

design the feedback controller

$$u = -Kx$$

Set the desired poles $J_1 = [\mu_1 = -2 + 4j, \mu_2 = -2 - 4j, \mu_3 = -10]$, $J_2 = 5 * J_1$, $J_3 = [\mu_1 = -2 + j, \mu_2 = -2 - j, \mu_3 = -5]$, and $J_4 = 2 * J_3$.

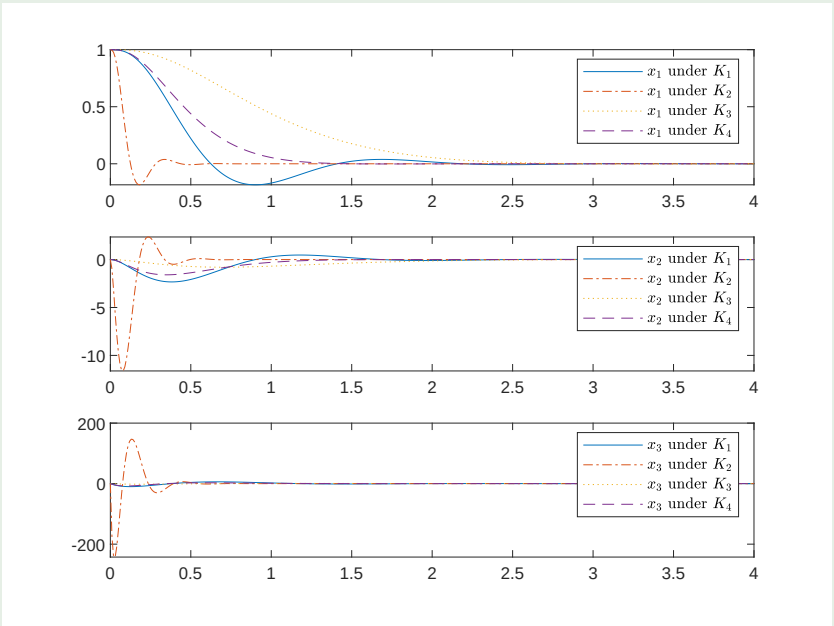


Figure 8: Responses with Dif. Feedback Gains

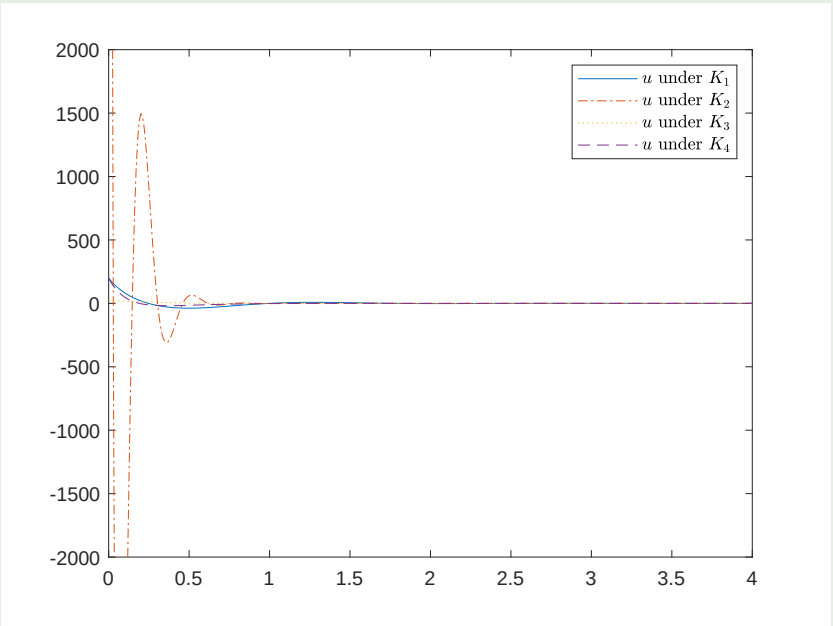


Figure 9: Inputs under Dif. Feedback Gains

Remark

Comparisons and discussions:

1. J_1 *v.s.* J_2, J_3 *v.s.* J_4*:*
   Both the real and imaginary parts of eigenvalues in J_1 *and* J_3 *are enlarged with varying ratios. The larger eigenvalue (real or imaginary part) leads to higher gain, faster response, and higher control input. Higher gain (large eigenvalue) is effective on speed up control and robustness, but also is with more input consumption.*
2. J_1 *v.s.* J_3, J_2 *v.s.* J_4*:*
   The desired eigenvalues with the same real part but different imaginary parts have a different response, like K_1 *and* K_3, K_2 *and* K_4*. Smaller imaginary parts will lead to lower control gains, lower overshot (response amplitude), and less control energy.*



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7 Stability Analysis based on Lyapunov Criterion

- Stability for the State-space Model



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Theorem

For a general homogeneous (autonomous or zero input) state equation model below

$$\dot{x} = f(x), x(0) = x_0 \tag{101}$$

we consider a general value functions with continuous partial derivatives in each state variable that are positive definite,

$$V(x) \begin{cases} = & 0, \text{ if } x = 0 \\ > & 0, \text{ if } x \neq 0 \end{cases} \tag{102}$$

whose derivative

$$\begin{aligned} \dot{V}(x) &= \sum_{i=1}^{n} \frac{\partial V}{\partial x_i} \dot{x}_i \\ &= \frac{\partial V}{\partial x} \dot{x} \\ &= \frac{\partial V}{\partial x} f(x) \end{aligned} \tag{103}$$

such, if $\dot{V}(x) < 0$, one can determine that the equilibrium state at the origin in system (101) is uniformly asymptotically stable.



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Theorem

Consider the following linear time invariant system

$$\dot{x} = Ax, \tag{104}$$

where x is a state vector, and $A \in \mathcal{R}^{n\times n}$ is an constant nonsingular matrix. (104) is asymptotically stable, if there exist positive definite matrices P and Q such that

$$A^T P + PA = -Q \tag{105}$$

Proof.

Select Lyapunov functional

$$V = x^T P x$$

where $P > 0$, such that

$$\begin{aligned} \dot{V} &= x^T P\dot{x} + \dot{x}^T P x \\ &= x^T (A^T P + P * A)x \\ &= x^T Q x \end{aligned}$$

therefore, if (105) holds, $\dot{V} < 0$. □

Example

Consider the system described below

$$\begin{aligned} \dot{x}_1 &= x_2 - x_1(x_1^2 + x_2^2) \\ \dot{x}_2 &= -x_1 - x_2(x_1^2 + x_2^2) \end{aligned}$$

Determine its stability.
Try selecting $V(x) = x_1^2 + x_2^2$, $V(x) = p_1 x_1^2 + p_2 x_2^2$

Example

Consider the system described by

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 - x_2 \end{aligned}$$

Determine its stability.
Assume there exists a positive matrix

$$V(x) = x^T P x$$

$P > 0$, and assume that

$$A^T P + PA = -I$$

Example

such that one can obtain the solution of P

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix} \tag{106}$$

It obtains that $P > 0$ and $A * P + P * A < 0$.



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8 Controller based on Observer

- Observer
- Observer-based Controller



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Consider the plant

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned} \tag{107}$$

The observer can be

$$\dot{\tilde{x}} = A\tilde{x} + Bu + L\left(y - C\tilde{x}\right) \tag{108}$$

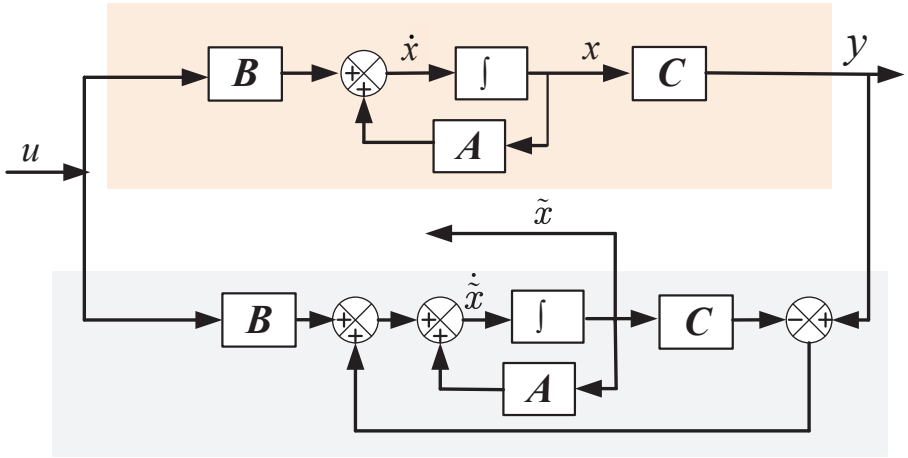


Figure 10: Full State Observer

$\tilde{x}$	the estimate of x
$C\tilde{x}$	the estimated output
L	observer gain that $\lim_{t\to\infty} \tilde{x} = x(t)$

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Remark

The observer, can not only estimate the unmeasurable part of the system state but also can smooth the given part. To some extent, it's also a filter.

Define the observer error $e = x - \tilde{x}$, with the system model and observer (108), one can obtain the error dynamics

$$\dot{e} = (A - LC)\, e \tag{109}$$

If matrix $A - LC$ is a stable matrix (Hurwitz, Negative Root, Left s Plane, Lyapunov stable,...) that the error state converges finally to zero for any initial state $e(0)$, i.e. $\tilde{x}$ approaches to $x(t)$ gradually.

$$\lim_{t \to \infty} \tilde{x}(t) = x(t) \tag{110}$$

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8 Controller based on Observer

- Observer
- Observer-based Controller

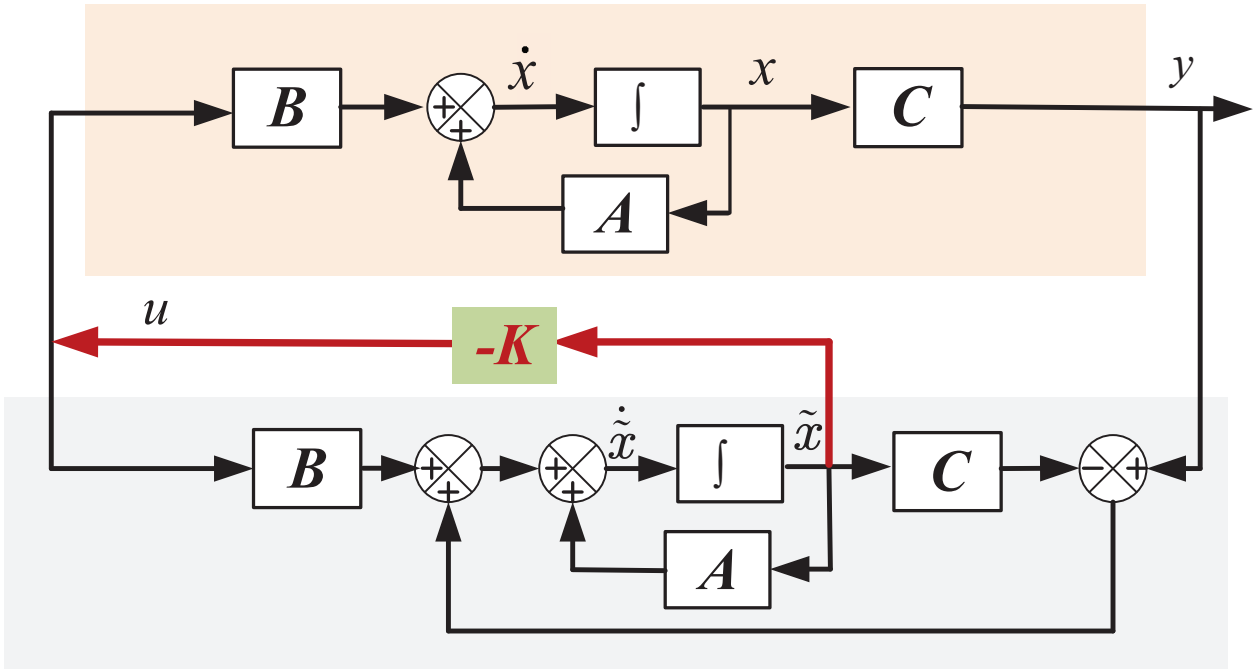


Figure 11: Observer-based Controller

The control input is designed with the observed system state $\tilde{x}$ rather than the original system state $x(t)$

$$u = -K\tilde{x}(t) \tag{111}$$

Substitute the observer-based control input (111) into (107)

$$\begin{cases} \dot{x} &= Ax - BK\tilde{x} \\ \dot{\tilde{x}} &= A\tilde{x} - BK\tilde{x} + L\left(y - C\tilde{x}\right) \end{cases} \tag{112}$$

and with $e = x - \tilde{x}$, one has that

$$\begin{cases} \dot{x} &= \left(A - BK\right)x + BKe \\ \dot{e} &= \left(A - LC\right)e \end{cases} \tag{113}$$

therefor, if one defines $\xi = \left[x^T \ e^T\right]^T$, it verifies that

$$\dot{\xi} = \bar{A}\xi \tag{114}$$

where

$$\bar{A} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix}$$

(114) determines the effectiveness of observer and controller together. If $\bar{A}$ is Hurwitz, with negative eigenvalues, then the system is stable (114).

Example

Consider the system below

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

design the observer-based feedback controller

$$u = -K\tilde{x}$$

the desired eigenvalues of $A - BK$ are

$$J_c = [\mu_1 = -2 + 4j, \mu_2 = -2 - 4j, \mu_3 = -10]$$

and the desired eigenvalues of $A - LC$ are

$$J_{O1} = [-2, -3, -4], J_{O2} = 5 * J_{O1}$$

Calculate the control and observer gains, and compare the system response, estimated system states under different observer gains.



Example

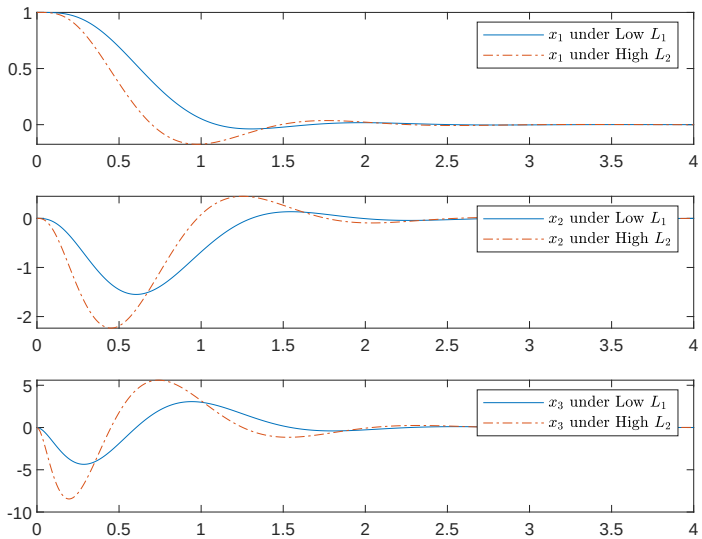
It can be obtained that

$$K = [199, 55, 8]$$

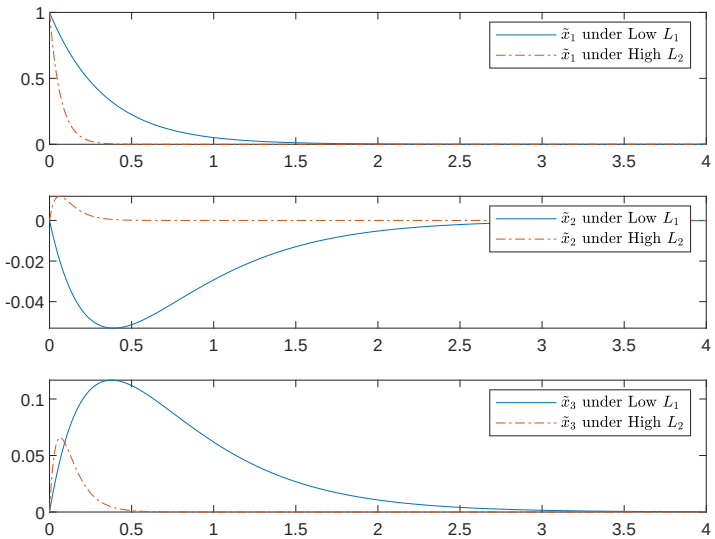
and

$$L_1 = \begin{bmatrix} 2.9887 & 1.0412 \\ 0.3751 & 0.0113 \\ -1.8548 & 2.9816 \end{bmatrix}, L_2 = \begin{bmatrix} 15.0203 & 0.7571 \\ -0.5421 & 23.9797 \\ -3.8017 & 50.9495 \end{bmatrix}$$

State Responses under Low and High Observer Gains



Observer State under Low and High Gains



Example

Same K but Different L:

- Higher L leads to faster responses of $\tilde{x}$
- Higher L leads to faster responses of x
- Higher L leads to higher consumption u

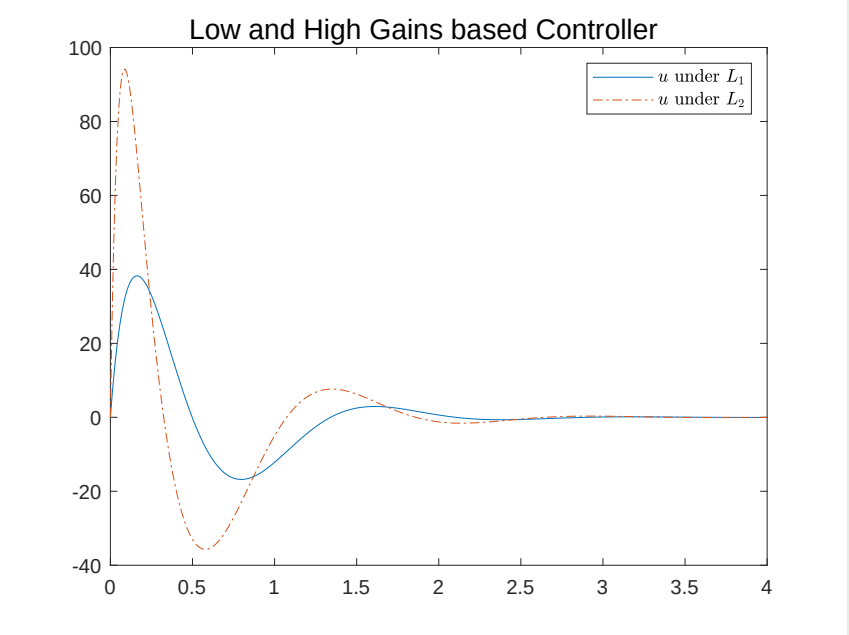


Figure 12: Control Inputs with Same Controllers but Different Observers

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1 Introduction

2 State-space Equation Model

3 State - space Representation of Transfer - function Systems

4 Solving the Time-invariant State Equation

5 Controllability and Observability of System

6 Pole Placement

7 Stability Analysis based on Lyapunov Criterion

8 Controller based on Observer

9 Chapter Conclusion

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- The advantages and motivations of using the state-space model: Simplified form, Useful for the classical system, Time-domain directly,. etc.;
- Why observability and controllability; How to determine the observability and controllability;
- Design controller for the state-space model system with the pole-placement method;
- Analyzing the system stability with general power criteria-Lyapunov method;
- Why observer? How to obtain the full system state when only a part state is measured;
- Observer-based controller design. And how to determine the effectiveness.



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谢谢!

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